CO-4 | 20/03/25

(Vector Calculus)

* **Scalar**:- A quantity which has ~~also~~ only magnitude.

Ex:- temp. mass, volume, speed, distance.

* **Vector**:- A quantity which have both magnitude & direction is called vector.

Ex:- Velocity, force, acceleration.

→ If $\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$, then

(i) $|\bar{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ (magnitude).

(ii) Unit vector in the direction of the vector $\bar{a}$ is taken as $\hat{a} = \frac{\bar{a}}{|\bar{a}|}$

* The dot product of two vectors is always scalar.

Ex:- $\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$

$\bar{b} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$

$\bar{a}\cdot\bar{b} = a_1b_1 + a_2b_2 + a_3b_3$ (Dot product)

$$\bar{a}\times\bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

(~~scalar~~)
(Cross Product)

→ If θ is angle inbetween two vectors

$\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$

$\bar{b} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$ then,

$$\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|\cdot|\bar{b}|}$$

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Scalar point function

If every point of R^3 is assigned to some scalar then that function f is called scalar point function.

Ex:-

$f(x,y,z) = xy + yz + zx$

$f(x,y,z) = x^2y^2 + xyz - 4.$

Vector valid function

If every point of R^3 is assigned with some vector, then that corresponding function $\bar{f}$ is called vector valid function.

Ex:-

$\bar{F}(x,y,z) = xy\bar{i} + yz\bar{j} + zx\bar{k}$

$\bar{f}(x,y,z) = x^2y^2\bar{i} - z^2\bar{j} + y^2\bar{k}$

Note:-

A vector valid function $\bar{f}(x,y,z)$ is represented as $\bar{f}(x,y,z) = f_1(x,y,z)\bar{i} + f_2(x,y,z)\bar{j} + f_3(x,y,z)\bar{k}$.

$$\boxed{\bar{f} = f_1\,\bar{i} + f_2\,\bar{j} + f_3\,\bar{k}}$$

Vector Differential operator

The vector differential operator ∇ (Del) is defined as

$$\boxed{\nabla = \bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}}$$

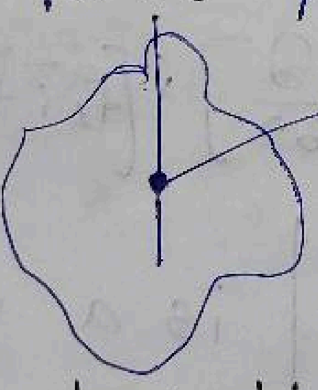
Gradient:-

The gradient of scalar point function $f(x,y,z)$ is defined as

$$\text{grad}(f) = \nabla f = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)$$

$$\boxed{\text{grad}(f) = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial F}{\partial z}}$$ is a vector.

Note:- The normal vector to the surface at the point 'p' on surface $f = (\nabla f)_{at\,p}$



$f(x,y,z)$

$\text{Normal} = (\nabla f)_{at\,P}$.

Directional derivative

The directional derivative of scalar point function $f(x,y,z)$ at the point $p(a,b,c)$ in the direction of the vector $\bar{a}$ is taken as

$$= (\nabla f)_{at\,p} \cdot \hat{a} \quad \text{where } \hat{a} = \frac{\bar{a}}{|\bar{a}|}$$

Divergence:-

The divergence of vector valid function $\bar{F} = f_1\hat{i} + f_2\hat{j} + f_3\hat{k}$ is defined as

$$div(\bar{F}) = \nabla\cdot\bar{F}$$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)\cdot\left(f_1\bar{i} + f_2\bar{j} + f_3\bar{k}\right)$$

$$\boxed{div(\bar{F}) = \nabla\cdot\bar{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}}$$

Note:- A vector valid function $\bar{F}$ is called Solenoidal if divergence $\bar{F} \Rightarrow div\,\bar{F} = \nabla\cdot\bar{F} = 0$.

Curl:-

The curl of vector valid function $\bar{f} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ is defined as

$$curl(\bar{F}) = \nabla\times\bar{F}$$

$$= \left[\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right]\times\left[f_1\bar{i} + f_2\bar{j} + f_3\bar{k}\right]$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$ is a vector.

Note:-

A vector valid function is called irrotational if $curl\cdot\bar{F} = \bar{0}$.

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① Find grad f, where $f = x^3 - y^3 + 3xyz$ at $P(1,2,3)$.

Sol: $\nabla f = \left[\bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}\right]$

$= \bar{i}\frac{\partial}{\partial x}(x^3 - y^3 + 3xyz) + \bar{j}\frac{\partial}{\partial y}(x^3 - y^3 + 3xyz) +$

$\bar{k}\frac{\partial}{\partial z}(x^3 - y^3 + 3xyz)$.

$= \bar{i}(3x^2 + 3yz) + \bar{j}(-3y^2 + 3xz) +$

$\bar{k}(3xy)$.

∇f at $P(1,2,3)$

$= \bar{i}(3(1)^2 + 3(2)(3)) + \bar{j}(-3(2)^2 + 3(1)(3)) +$

$\bar{k}(3(1)(2))$

$= \bar{i}(3 + 18) + \bar{j}(-12 + 9) + \bar{k}(6)$

$= 21\bar{i} - 3\bar{j} + 6\bar{k}$ //

② Determine unit normal vector to the surface $x^2y - 2xz = 4$ at the point $(1, -1, 0)$.

Sol:- normal vector to the surfaces also known as grad $f = \nabla \cdot f$

$= \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$.

$= \bar{i}\frac{\partial}{\partial x}[x^2y-2xz]+\bar{j}\frac{\partial}{\partial y}(x^2y-2xz]+\bar{k}\frac{\partial}{\partial z}(x^2y-2xz)$

$= \bar{i}(2xy-2z]+\bar{j}[x^2]+\bar{k}[-2x]$

$\bar{n} = i[2xy-2z]+\bar{j}[x^2]+\bar{k}[-2x]$

∇f at $p(1,-1,0)$

$i[2(1)(-1)-2(0)]+\bar{j}[1]^2-\bar{k}[-2(1)]$

$$\boxed{\bar{n} = -2\bar{i}+\bar{j}-2\bar{k}}$$

Unit normal vector $= \frac{\bar{n}}{|\bar{n}|}$ $= |\bar{n}| = \sqrt{4+1+4}$

$= \sqrt{9} = 3$ //.

$\therefore \bar{n} = \frac{-2\bar{i}+\bar{j}-2\bar{k}}{3}$

③ Determine the directional derivative of scalar point of function $f = xy^2 - y^3z - z^3x$ in the direction of the vector $\bar{i}+\bar{j}+2\bar{k}$ at $P(2,1,3)$.

Sol:- $f = xy^2 - y^3z - z^3x$

$\bar{a} = \bar{i}+\bar{j}+2\bar{k}$

$\nabla f = \bar{i}\frac{\partial}{\partial x}(xy^2-y^3z-z^3x)+\bar{j}\frac{\partial}{\partial y}(xy^2-y^3z-z^3x)$

$+\bar{k}\frac{\partial}{\partial z}(xy^2-y^3z-z^3x)$

$= \bar{i}(y^2-z^3)+\bar{j}(2xy-3y^2z)+\bar{k}(-y^3-3z^2x)$

∇f at $P(2,1,3)$

$+\bar{k}\frac{\partial}{\partial z}$
$x^2y - 2xz)$
$2xz]$
$-2x]$.

$\bar{k}[-2(1)]$

$+1+4$

$=3$ //.

ve of
$y^3z -$
ector

$z^3x)$

$3z^2x)$

$\bar{i}(1^3(3)^3) + \bar{j}(2(2)(1) - 3(1)^2 3) + \bar{k}(-(1)3 + 3(3)^2 2)$

$= \bar{i}[-26] + \bar{j}(4-9) + \bar{k}(-9 - 54]$

$= -(26\bar{i} + 5\bar{j} + 53\bar{k})$

$\hat{a} = \frac{\bar{a}}{|\bar{a}|} = \frac{\bar{i}+\bar{j}+2\bar{k}}{\sqrt{1+1+4}} = \frac{\bar{i}+\bar{j}+2\bar{k}}{\sqrt{6}}$

$= -(26\hat{i} + 5\hat{j} + 53\bar{k}) \cdot \frac{\bar{i}+\bar{j}+2\bar{k}}{\sqrt{6}}$

$= \frac{-26-5-110}{\sqrt{6}} = \frac{-141}{\sqrt{6}}$ //.

(4) Evaluate directional derivative of $\phi = xy - yz - zx$ at A in the direction of vector $\vec{AB}$, when $A = (1,2,0)$, $B = (1,0,3)$

Sol: Vector $\vec{AB} = \vec{OB} - \vec{OA}$

$= (\bar{i} + 0\bar{j} + 3\bar{k}) - (\bar{i} + 2\bar{j} + 0\bar{k})$

$= -2\bar{j} + 3\bar{k}$

$\nabla f = \bar{i}\frac{\partial}{\partial x}(xy - yz - zx) + \bar{j}\frac{\partial}{\partial y}(xy - yz - zx)$

$+ \bar{k}\frac{\partial}{\partial z}(xy - yz - zx)$

$= \bar{i}[y - z] + \bar{j}[x - z] + \bar{k}[y - x]$ At $(1,2,0)$

$= (\bar{i} + \bar{j} - 3\bar{k}) \cdot \frac{(-2\bar{j} + 3\bar{k})}{\sqrt{13}} = \nabla f \cdot \hat{a}$

$\hat{a} = \frac{\bar{a}}{|\bar{a}|}$

$= \frac{-2\bar{i} + 3\bar{k}}{\sqrt{4+9}}$

$= \frac{-2\bar{i} + 3\bar{k}}{\sqrt{13}}$

$= \frac{2(0) + 1(-2) - 3(3)}{\sqrt{13}}$

$= \frac{-2-9}{\sqrt{13}} = \frac{-11}{\sqrt{13}}$ //.

6) Find angle in between surface $x^2-y^2-z^2=4$ & $x^2-y^2-z=2$ at the point $(0,-1,+2)$.

Sol:- The angle in between surfaces is angle in between their normals.

Take $\phi = x^2-y^2-z^2-4$ at point
$\psi = x^2-y^2-z-2$ $(0,-1,2)$

Take $\bar{a} = (\nabla\phi)$ at P.
$\bar{b} = (\nabla\psi)$ at P

Grad$(\nabla)\phi$

$$\nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}[x^2-y^2-z^2-4] + \bar{j}\frac{\partial}{\partial y}[x^2-y^2-z^2-4] + \bar{k}\frac{\partial}{\partial z}[x^2-y^2-z^2-4].$$

$$= \bar{i}[2x] + \bar{j}[-2y] + \bar{k}[-2z]$$

Now $\bar{a} = \nabla\phi$ at P.

$\bar{a} = 2\bar{j} - 4\bar{k}$.

$(\nabla\psi)$

$$\nabla\psi = \bar{i}\frac{\partial\psi}{\partial x} + \bar{j}\frac{\partial\psi}{\partial y} + \bar{k}\frac{\partial\psi}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}[x^2-y^2-z-2] + \bar{j}\frac{\partial}{\partial y}[x^2-y^2-z-2] + \bar{k}\frac{\partial}{\partial z}[x^2-y^2-z-2]$$

$$= \bar{i}[2x] + \bar{j}[-2y] + \bar{k}[-1]$$

$\nabla\psi$ at $P(0,-1,2)$

$\bar{b} = +2\bar{j} - \bar{k}$ //

Let θ be angle in between $\bar{a}$ & $\bar{b}$

$$\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$$

$$|\bar{a}| = \sqrt{(2)^2+(-4)^2} = \sqrt{20}$$

$$|\bar{b}| = \sqrt{(2)^2+(-1)^2} = \sqrt{5}$$

$$\frac{(2\bar{j}-4\bar{k})\cdot(2\bar{j}-\bar{k})}{\sqrt{20}\cdot\sqrt{5}} \Rightarrow \frac{2\cdot 2+(-4)(-1)}{\sqrt{100}}$$

$$= \frac{8}{\sqrt{100}} = \frac{8}{10} = \frac{4}{5} //$$

$$\cos\theta = \frac{4}{5}$$

$$\boxed{\theta = \cos^{-1}\frac{4}{5}}$$

Q. Determine the angle in between the normal to the surface $xy^2z = 3x - z^2$ at the points $P(2,0,2)$ & $Q(1,-1,1)$

Sol:- Take $f = xy^2z - 3x + z^2$, $P(2,0,2)$, $Q(1,-1,1)$

Take $\bar{a} = (\nabla f)$ at $P(2,0,2)$

$\bar{b} = (\nabla f)$ at $Q(1,-1,1)$.

$$\bar{a} = \bar{i}\frac{\partial f}{\partial x}[xy^2z - 3x + z^2] + \bar{j}\frac{\partial f}{\partial y}[xy^2z - 3x + z^2]$$

$$+ \bar{k}\frac{\partial f}{\partial z}[xy^2z - 3x + z^2]$$

$$= \bar{i}[y^2z - 3] + \bar{j}[xz(2y)] + \bar{k}[xy^2 + 2z]$$

∇f at $P(2,0,2)$

$$\bar{a} = \bar{i}[-3] + \bar{j}[0] + \bar{k}[+4] \Rightarrow -3\bar{i} + 4\bar{k}$$

$\bar{b} = (\nabla f)_{at}\ Q(1,-1,1)$

$= -2\bar{i} - 2\bar{j} + 3\bar{k}.$

$\cos\theta = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{a}|\cdot|\bar{b}|}$; $|\bar{a}| = \sqrt{9+16} = 5$

$|\bar{b}| = \sqrt{4+4+9} = \sqrt{17}$

$\dfrac{(-3\bar{i} + 4\bar{k})\cdot(-2\bar{i} - 2\bar{j} + 3\bar{k})}{5\cdot\sqrt{17}} \Rightarrow \dfrac{(-3)(-2) + 0 + 12}{5\sqrt{17}}$

$= \dfrac{18}{5\sqrt{17}}$ //.

Note:- $\text{div}\,\bar{f} = \nabla\cdot\bar{F} = \dfrac{\partial f_1}{\partial x} + \dfrac{\partial f_2}{\partial y} + \dfrac{\partial f_3}{\partial z}$

$\bar{f} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$\text{curl}\,\bar{f} = \nabla\times\bar{f} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$

① If $\bar{f} = x^2y\bar{i} + y^2z\bar{i} + z^2x\bar{k}$, $\text{div}(\bar{F})$ at $P(1,1,2)$

sol:- $\bar{f} = x^2y\bar{i} + y^2z\bar{j} + z^2x\bar{k}$.

$= f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ $\quad P(1,1,2)$

$\text{div}\,\bar{F} = \dfrac{\partial f_1}{\partial x} + \dfrac{\partial f_2}{\partial y} + \dfrac{\partial f_3}{\partial z}$

$= \dfrac{\partial}{\partial x}[x^2y] + \dfrac{\partial}{\partial y}[y^2z] + \dfrac{\partial}{\partial z}[z^2x]$

$= y(2x) + z(2y) + x(2z) \Rightarrow 2xy + 2yz + 2zx$

div $\bar{F}$ at $P(1,1,2)$

$2xy + 2yz + 2zx = 2(1)(1) + 2(1)(2) + 2(2)(1)$

$= 2 + 4 + 4$

$= 10$ //

② Evaluate div $\bar{F}$ & curl $\bar{F}$ at $(2,-1,3)$ where $\bar{F} = xyz\,\bar{i} - 2x^2y\,\bar{j} - 3y^2z\,\bar{k}$.

$f_1 = xyz,\ f_2 = 2x^2y,\ f_3 = 3y^2z$

$\bar{F} = \frac{\partial}{\partial x}[xyz] - \frac{\partial}{\partial y}[2x^2y] - \frac{\partial}{\partial z}[3y^2z]$

$= [yz] - 2x^2(1) - 3y^2(1)$

$\bar{F} = [yz] - [2x^2] - [3y^2]$

$= -3 - (8) - 3$

$= -14$ //.

~~$\bar{F} = yz$~~

~~div $\bar{F}$ at $P(2,-1,3)$~~

~~$\bar{F} = (-1)3$~~

~~$\boxed{\bar{F} = -3}$ //~~

curl $\bar{F} = \nabla \times \bar{F}$

$$\begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & -2x^2y & -3y^2z \end{vmatrix}$$

$= \bar{i}\left[\frac{\partial}{\partial y}(-3y^2z) - \frac{\partial}{\partial z}(-2x^2y)\right] - \bar{j}\left[\frac{\partial}{\partial x}(3y^2z) - \frac{\partial}{\partial z}(xyz)\right]$.

$+ \bar{k}\left[\frac{\partial}{\partial x}(-2x^2y) - \frac{\partial}{\partial y}(-xyz)\right]$.

$= \bar{i}[-6zy - 0] - \bar{j}[0 - xy] + \bar{k}[-4xy - xz]$

At $P(2,-1,3)$

$= -6(3)(-1)\bar{i} + 2(-1)\bar{j} - \bar{k}\left(4(2)(-1)+2(3)\right)$

$= 18\bar{i} - 2\bar{j} + 2\bar{k}$ //

Q. Determine a, b, c values if $\bar{F} = (2x+3y+az)\bar{i} + (bx+2y+3z)\bar{j} + (2x+cy+3z)\bar{k}$ is irrotational.

Sol:-

$f_1 = 2x+3y+az$

$f_2 = bx+2y+3z$

$f_3 = 2x+cy+3z$

$$\text{Curl } \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (2x+3y+az) & (bx+2y+3z) & (2x+cy+3z) \end{vmatrix}$$

$$\bar{i}\left[\frac{\partial}{\partial y}(2x+cy+3z) - \frac{\partial}{\partial z}(bx+2y+3z)\right] - \bar{j}\left[\frac{\partial}{\partial x}(2x+cy+3z) - \frac{\partial}{\partial z}(2x+3y+az)\right]$$

$$+\bar{k}\left[\frac{\partial}{\partial x}(bx+2y+3z) - \frac{\partial}{\partial y}(2x+3y+az)\right]$$

$$\bar{i}[c-3] - \bar{j}[2-a] + \bar{k}[b-3]$$

$$[c-3]\bar{i} - [2-a]\bar{j} + [b-3]\bar{k} = 0 \quad //$$

$c-3=0 \;|\; 2-a=0 \;|\; b-3=0$

$c=3 \;|\; a=2 \;|\; b=3$ //

Q. Determine $p = ?$, if $\bar{f} = (x+3y)\bar{i} + (y-2z)\bar{j} + (x+pz)\bar{k}$ is solenoidal vector.

Sol:- $\operatorname{div} \bar{f} = 0$

$$= \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} = 0$$

$$= \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+pz) = 0$$

$$= 1 + 1 + p = 0$$

$$\boxed{p = -2}$$

Vector Integration

Line Integral

* The line integral of the vector valued function $\bar{f} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ along the curve 'C' is defined as $\int_C \bar{f}\cdot d\bar{r}$

where $\bar{f} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$$\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$$

$$d\bar{r} = \bar{i}\,dx + \bar{j}\,dy + \bar{k}\,dz$$

$$\int_C \bar{f}\cdot d\bar{r} = \int_C (f_1\bar{i} + f_2\bar{j} + f_3\bar{k})\cdot(\bar{i}\,dx + \bar{j}\,dy + \bar{k}\,dz)$$

$$\boxed{\int_C \bar{f}\cdot d\bar{r} = \int_C f_1\,dx + f_2\,dy + f_3\,dz}$$

① Evaluate $\int_C \bar{F}\cdot d\bar{r}$ where $\bar{F} = 2xy\bar{i} - 3yz\bar{j} + 4xz\bar{k}$ along C: from (0,0,0) to (1,1,1) with $x = t, y = t^2$ & $z = t^3$

Sol:- $f_1 = 2xy, f_2 = 3yz, f_3 = 4xz$.

$C: x = t, y = t^2, z = t^3$ (0,0,0) to (1,1,1).

$$\int_C \bar{f}\cdot d\bar{r} = \int_C f_1\,dx + f_2\,dy + f_3\,dz$$

$$= \int_C 2xy\,dx - 3yz\,dy + 4xz\,dz$$

For $x = t \Rightarrow dx = dt$

$y = t^2 \Rightarrow dy = 2t\,dt$

$z = t^3 \Rightarrow dz = 3t^2\,dt$

$$\int_C 2(t)(t^2)\,dt - 3(t^2)(t^3)\,2t\,dt + 4(t)(t^3)\,3t^2\,dt$$

$t: 0 \to 1$

$$\int_0^1 2t^3\,dt - 6t^6\,dt + 12\,t^6\,dt$$

$$= 2\left[\frac{t^4}{4}\right]_0^1 - 6\left[\frac{t^7}{7}\right]_0^1 + 12\left(\frac{t^7}{7}\right)_0^1$$

$$= \cancel{2}\left[\frac{1}{4}\right] - 6\left[\frac{1}{7}\right] + 12\left[\frac{1}{7}\right] = \frac{1}{2} - \frac{6}{7} + \frac{12}{7}$$

$\frac{1}{2} + \frac{6}{7} = \frac{7+12}{14} = \frac{19}{14}$ //.

Q. Evaluate $\boxed{\text{workdone by the force } \bar{f} = \text{Line integral.}}$

$\bar{f} = 2xy\,\bar{i} - 3z\,\bar{j} + 5x\,\bar{k}$ along the curve $C: x = t^2+1 \quad t: 0 \to 1.$

$y = 2t^2$

$z = t^3$

Sol:- The Line integral is

$$\int_C \bar{F}\cdot d\bar{r} = \int_C f_1\,dx + f_2\,dy + f_3\,dz$$

$$= \int_C 2xy\,dx - 3z\,dy + 5x\,dz$$

$$= \int_C 2(t^2+1)(2t^2)\,2t\,dt - 3(t^3)\,4t\,dt + 5(t^2+1)\,3t^2\,dt$$

$$= \int_0^1 8(t^5+t^3)\,dt - 12t^4 + 15(t^4+t^2)\,dt$$

$$= 8\left[\frac{t^6}{6} + \frac{t^4}{4}\right]_0^1 - 12\left[\frac{t^5}{5}\right]_0^1 + 15\left[\frac{t^5}{5} + \frac{t^3}{3}\right]_0^1$$

$$= 8\left[\frac{1}{6} + \frac{1}{4}\right] - 12\left[\frac{1}{5}\right] + 15\left[\frac{1}{5} + \frac{1}{3}\right]$$

$$= 8\left[\frac{2+3}{12}\right] - \frac{12}{5} + \cancel{15}\left[\frac{3+5}{1\cancel{5}}\right].$$

$\frac{40}{12} - \frac{12}{5} + 8 = \frac{200 - 144 + 480}{\cancel{60}}$

$= \frac{5\cancel{3}6}{160} =$

$\frac{10}{3} - \frac{12}{5} + 8$

$= \frac{134}{15}$ //.

$\boxed{02/04/25}$

Q. Evaluate workdone by the force $\bar{F} = 2x^2\bar{i} + (4xz - y)\bar{j} + 2z\bar{k}$ when it moves a particle along the straight line passing $(0,0,0)$ & $(2,1,3)$.

Sol:- $\int_C \bar{F} \cdot d\bar{r} = \int_C f_1 dx + f_2 dy + f_3 dz$

$= \int_C 2x^2 dx + (4xz - y) dy + 2z dz$

C is straight line from $(0,0,0)$ to $(2,1,3)$.

So, $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$

$= \frac{x - 0}{2 - 0} = \frac{y - 0}{1 - 0} = \frac{z - 0}{3 - 0} = t$

$\frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t \Rightarrow x = 2t$

$y = t$

$z = 3t$

$dx = 2$

$dy = 1$

$dz = 3$

As $y = t$, we know that y varies from 0 to 1 from the given points so 't' varies from 0 to 1.

$t : 0 \to 1$

$$\int_C \bar{F}\cdot d\bar{r} = \int_C 2x^2\,dx + (4xz - y)\,dy + 2z\,dz$$

$$= \int_0^1 2(2t)^2\,2 + (4(2t)(3t) - t)\,1 + 2(3t)\,3$$

$$= \int_0^1 16t^2 + 24t^2 - t + 18t$$

$$= \int_0^1 16t^2 + 24t^2 - t + 18t \Rightarrow 16\left[\frac{t^3}{3}\right]_0^1 + 4\left[\frac{t^2}{2}\right]_0^1$$

$$= \frac{16}{3} + \frac{4}{2}$$

$$= \frac{32 + 12}{6} = \frac{44}{6}$$

$$= \frac{22}{3}$$

$$= \int_0^1 40t^2 + 17t$$

$$= 40\left[\frac{t^3}{3}\right]_0^1 + 17\left[\frac{t^2}{2}\right]_0^1$$

$$= \frac{40}{3} + \frac{17}{2} = \frac{80 + 51}{6} = \frac{131}{6}$$

Green's Theorem

→ Applicable only for xy plane.

In xy plane $\int_C m(x,y)dx + N(x,y)dy$

$$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial m}{\partial y}\right) dx\,dy$$ where $m(x,y)$, $N(x,y)$, $\frac{\partial N}{\partial x}$ & $\frac{\partial m}{\partial y}$ are continuous functions on the region 'R' bounded by closed curve 'C'.

⇒ Green's Theorem ⇒ $\int_C m\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial m}{\partial y}\right) dx\,dy$

Session - 27/28

① Use Green's Theorem $\int_C (xy - y^2)\,dx - x^2\,dy$, where 'C' is bounded by $y = x$ & $y = x^2$

Sol:- W.K.T $\int_C m\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial m}{\partial y}\right) dx\,dy$

Given $\int_C (xy - y^2)\,dx - x^2\,dy$

$M = (xy - y^2)$

$N = -x^2$

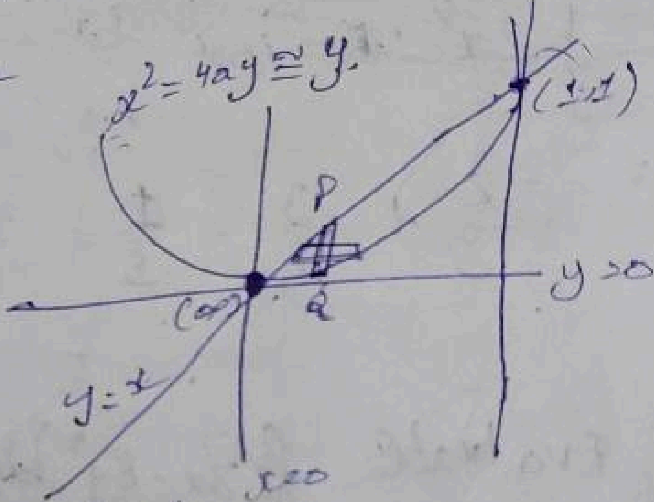
$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(xy - y^2) = x - 2y$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial y}(-x^2) = -2x$$

$$= \iint_R \left(-2x - (x-2y)\right) dx\, dy$$

$$= \iint_R (-3x+2y)\, dx\, dy$$

$C: y \to x$ & $y = x^2$



$y: x^2 \to x$

$x: 0 \to 1$

$y = x$ & $x^2 = y$

$x^2 = x$

$x^2 - x = 0$

$x[x-1] = 0$

$x = 0;\ x = 1$

$y = 0;\ y = 1$

$$I = \int_{x=0}^{1}\int_{y=x^2}^{x} (-3x+2y)\, dy\, dx$$

$$= \int_0^1 -3x\left[y\right]_{x^2}^{x} + 2\left[\frac{y^2}{2}\right]_{x^2}^{x} dx$$

$$= \int_0^1 -3x\left[x - x^2\right] + 2\left[\frac{x^2}{2} - \frac{x^4}{2}\right] dx$$

$$\int_0^1 \left(-3x^2 + 3x^3 + \cancel{x}\frac{(x^2-x^4)}{\cancel{x}}\right) dx$$

$$\int_0^1 \left[-3x^2 + 3x^3 + x^2 - x^4\right] dx$$

$$= \int_0^1 \left[-2x^2 + 3x^3 - x^4\right] dx$$

$$= -2\left[\frac{x^3}{3}\right]_0^1 + 3\left[\frac{x^4}{4}\right]_0^1 - \left[\frac{x^5}{5}\right]_0^1$$

$$= \frac{-2}{3} + \frac{3}{4} - \frac{1}{5} = \frac{-40+45-12}{60}$$

$$= \frac{-7}{60}$$

② Evaluate $\int_C (3x-8y^2)dx + (4y-6xy)dy$ where 'c' is the region bounded by $x=0, y=0, x+y=1$.

Sol:- W.K.T

$$\int_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dy\,dx$$

$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(4y-6xy)$ | $\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(3x-8y^2)$

$\frac{\partial N}{\partial x} = -6y$ | $\frac{\partial M}{\partial y} = -16y$

$]_0^1 + 3\left[\frac{x^4}{4}\right]_0^1$

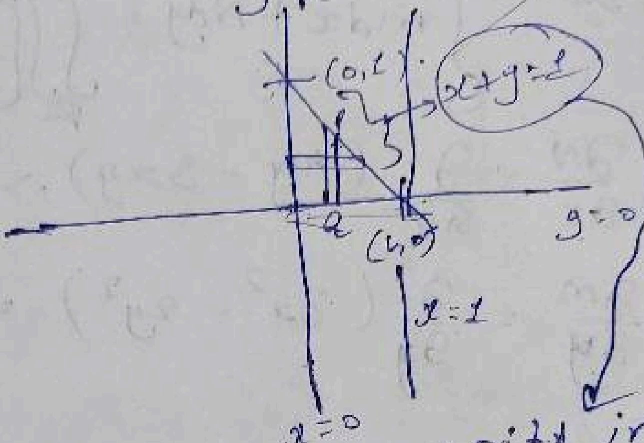
$]_0^1$

$)dy.$

ded

$$= \iint_R -6y - (-16y)\, dy\, dx \Rightarrow \iint_R (-6+16y)\, dy\, dx$$

$$= \iint_R (10y)\, dy\, dx \Rightarrow 10 \iint_R y\, dy\, dx.$$



write in intercept form

$$\frac{x}{1} + \frac{y}{1} = 1$$

$y: 0 \to 1-x$

$x: 0 \to 1$

$$10 \int_{x=0}^{1} \int_{y=0}^{1-x} y\, dy\, dx$$

$$10 \int_{x=0}^{1} \left[\frac{y^2}{2}\right]_0^{1-x} dx$$

$$\frac{10}{2} \int_{x=0}^{1} (1-x)^2\, dx \Rightarrow 5 \int_{x=0}^{1} \left[1 + x^2 - 2x\right] dx$$

$$= 5 \left[x\right]_0^1 + \left[\frac{x^3}{3}\right]_0^1 - 2\left[\frac{x^2}{2}\right]_0^1$$

$$= 5 \cdot [1] + \frac{1}{3} - 1$$

$$\frac{5}{1} + \frac{1}{3} - \frac{1}{1} = \frac{15+1-3}{3} = \frac{13}{3}$$

$$= 5 \int_{x=0}^{1} \left[(1-x)^2 - 0^2\right] dx$$

$$= 5 \left[\frac{\frac{(1-x)^3}{3}}{(-1)}\right]_0^1 =$$

$$= -\frac{5}{3} \left[(1-x)^3\right]_0^1 \Rightarrow +\frac{5}{3}$$

③ Apply Green's Theorem

$$\int_C (3x^2 - 2y^2)\,dx + (4y - 3xy)\,dy \quad C: y = \sqrt{x} \;\&\; y = x^2$$

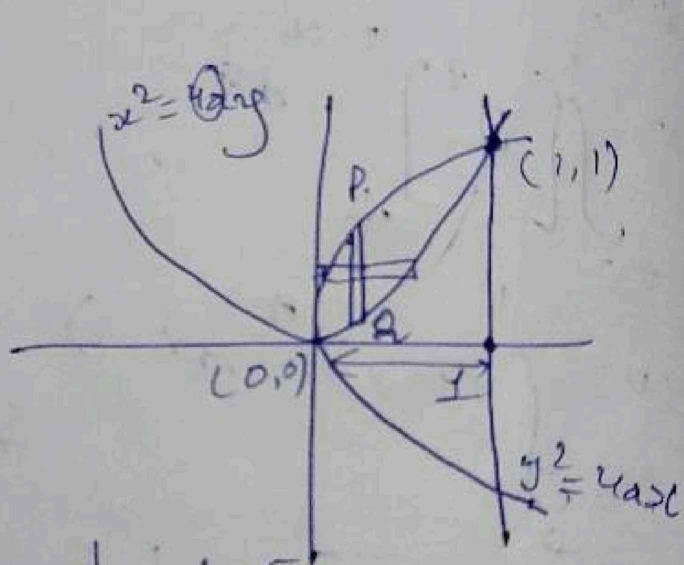
Sol:- $$\int_C M\,dx + N\,dy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right] dx\,dy$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(4y - 3xy) \Rightarrow -3y$$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(3x^2 - 2y^2) \Rightarrow -4y$$

$$= \iint_R [-3y - (-4y)]\,dx\,dy$$

$$= \iint_R y\,dx\,dy$$



$y = \sqrt{x}$ | $y = x^2$

$x = y^2$

$\therefore y : x^2 \to \sqrt{x}$

$x : 0 \to 1$

As $y^2 = x \Rightarrow y^4 = x^2 = y$

$\therefore y^4 - y = 0$

$y[y^3 - 1] = 0$

$y = 0 \mid y = 1$

$\Downarrow \qquad \Downarrow$

$x = (0,0) \quad (1,1)$

$$\int_{x=0}^{1}\int_{y=x^2}^{\sqrt{x}} y\,dy\,dx$$

$$= \int_{x=0}^{1}\left[\frac{y^2}{2}\right]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 \left[\frac{x}{2} - \frac{x^4}{2}\right] dx$$

$$\int_0^1 \frac{x-x^4}{2}\,dx \Rightarrow \frac{1}{2}\int_0^1 [x-x^4]\,dx$$

$$=\frac{1}{2}\left[\frac{x^2}{2}\right]_0^1-\left[\frac{x^5}{5}\right]_0^1$$

$$=\frac{1}{2}\times\left[\frac{1}{2}-\frac{1}{5}\right] \Rightarrow \frac{1}{2}\left[\frac{5-2}{10}\right]=\frac{1}{2}\times\frac{3}{10}$$

$$=\frac{3}{20}$$ //

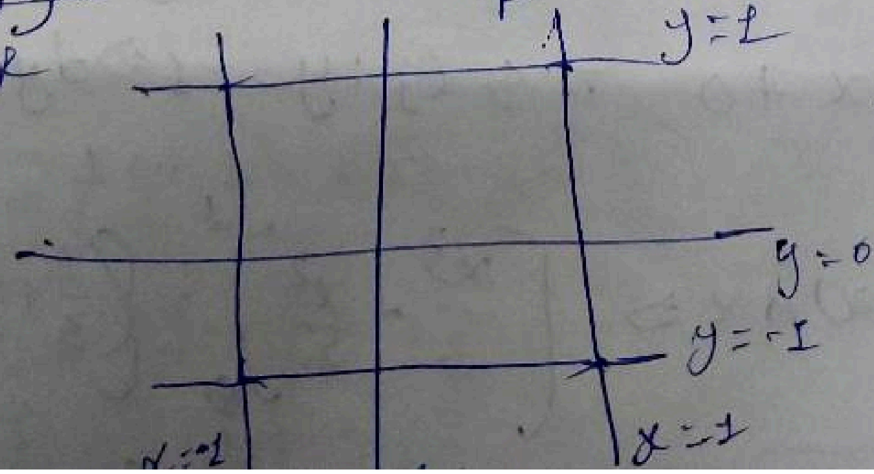
(4) Evaluate: $\int_C (x^2-xy)\,dx+(x^2-y^2)\,dy$, where C is the square formed by $x=\pm 1$ & $y=\pm 1$

Sol:- $$\int_C M\,dx+N\,dy=\iint_R\left[\frac{\partial N}{\partial x}-\frac{\partial M}{\partial x}\right]dx\,dy$$

$$\frac{\partial N}{\partial x}=\frac{\partial}{\partial x}(x^2-y^2) \Rightarrow 2x$$

$$\frac{\partial M}{\partial x}=\frac{\partial}{\partial y}(x^2-xy) \Rightarrow -x$$

$$\cancel{\iint_R \ldots (-x)} \quad \iint_R [2x-(-x)]\,dx\,dy$$



$x : -1 \to 1$

$y : -1 \to 1$

$$\iint_R [3x]\, dx\, dy \Rightarrow 3\iint_R x\, dx\, dy$$

$$3\int_{x=-1}^{1}\int_{y=-1}^{1} x\, dy\, dx \Rightarrow \cancel{\int_{x=-1}^{1}\left[\frac{x^2}{2}\right]_{-1}^{1} dx}$$

$$= \cancel{\int_{-1}^{1}\left[\frac{1}{2} - \frac{1}{2}\right] dx} \Rightarrow \cancel{\int_{-1}^{1} 0\, dx}$$

$$3\int_{-1}^{1} x\left[\int_{-1}^{1} 1\, dy\right] dx \Rightarrow 3\int_{-1}^{1} x\,[y]_{-1}^{1}\, dx$$

$$= 3\int_{-1}^{1} x(1+1)\, dx \Rightarrow 3\times 2\int_{-1}^{1} x\, dx \Rightarrow 6\left[\frac{x^2}{2}\right]_{-1}^{1}$$

$$= 6\left[\frac{1}{2} - \frac{1}{2}\right] = 0$$

(or)

<u>without</u> <u>applying</u> <u>green's</u> <u>theorem</u>

$$\int_C (x^2 - xy)\, dx + (x^2 - y^2)\, dy = \int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4}$$

$$\int_{C_1} (x^2 + x)\, dx + 0 \quad \text{As } C_1 : y = -1 \Rightarrow dy = 0$$

$$x : -1 \to 1$$

$$= \int_{x=-1}^{1} (x^2 + x)\, dx \Rightarrow \left[\frac{x^3}{3} + \frac{x^2}{2}\right]_{-1}^{1} = \left(\frac{1}{3} + \frac{1}{2}\right) - \left(\frac{1}{3}\right.$$

C_2

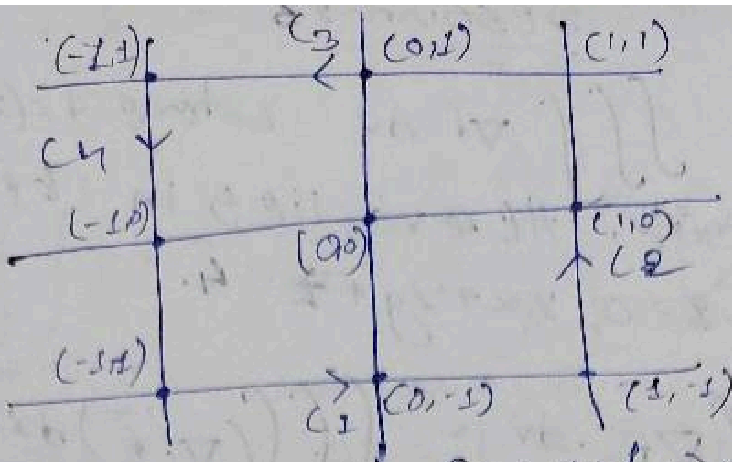
y

C_4

②

sol:

a



$C_2 : x = 1 \quad dx = 0$

$y : -1 \to 1$

$C_3 : y = 1 \Rightarrow dy = 0$

$x : 1 \to -1$

$C_4 :- \quad x = -1 \Rightarrow dx = 0$

$y : 1 \to -1$

$\boxed{09/04/25}$

Session-23/24

② The Temperature at the point $P(x,y,z)$ in the space is given by $T(x,y,z) = x^2 + y^2 - z$. ~~The temperature~~ A mosquito located at $(1,-1,2)$ desires to fly in such a direction that it gets cooled faster. Determine the direction in which the mosquito should fly.

Sol:- The mosquito need to fly in normal vector direction 'T' for cool as early as possible.

$$(\nabla T)_{(1,-1,2)} = \bar{i}\frac{\partial T}{\partial x} + \bar{j}\frac{\partial T}{\partial y} + \bar{k}\frac{\partial T}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}(x^2+y^2-z) + \bar{j}\frac{\partial}{\partial y}(x^2+y^2-z) + \bar{k}\frac{\partial}{\partial z}(x^2+y^2-z)$$

$$= \bar{i}\cdot 2x + \bar{j}\cdot 2y + \bar{k}(-1)$$

$$= 2x\bar{i} + 2y\bar{j} - \bar{k}.$$

Session-26

Evaluate $\iiint \nabla \bar{F}\, dv$ where $\bar{F} = (2x^2 - 3z)\bar{i} - 2xy\bar{j} - 4x\bar{k}$. Here 'v' lies in between $x = 0, y = 0, z = 0, 2x + 2y + z = 4$.

Sol:- $\iiint_V \nabla \bar{F} \cdot dv = \iiint_V (\nabla \cdot \bar{F})\, dx\, dy\, dz$

$\bar{F} = (2x^2 - 3z)\bar{i} - 2xy\bar{j} - 4x\bar{k}$

$f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$\nabla \bar{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$

$= \frac{\partial}{\partial x}(2x^2 - 3z) + \frac{\partial}{\partial y}(-2xy) + \frac{\partial}{\partial y}(-4x)$

$\nabla F = 4x - 2x + 0 \Rightarrow 2x$

$I = \iiint_V (2x)\, dx\, dy\, dz$

Given, $x = 0, y = 0, z = 0, 2x + 2y + z = 4$

$z = 4 - 2x - 2y$

$z : 0 \rightarrow 4 - 2x - 2y$

$z = 0 \Rightarrow 2x + 2y + z = 4$

$2x + 2y = 4$

$\cancel{2}(x + y) = \cancel{4}\, 2$

$\boxed{y = 2 - x}$

$y: 0 \to 2-x$ | $x+y=2$ $(z=0)$

$x: 0 \to 2$ | $\boxed{y=2}$

$$\int_{z=0}^{4-2x-2y}\int_{y=0}^{2-x}\int_{x=0}^{2} 2x \, dx \, dy \, dz \Rightarrow \int_{z=0}^{4-2x-2y}\int_{y=0}^{2-x} 2\left[\frac{x^2}{2}\right]_0^2 dy\,dz$$

$$= \int_0^{4-2x-2y}\int_0^{2-x} 4 \, dy \, dz \Rightarrow \int_0^{4-2x-2y} 4[y]_0^{2-x} \, dz$$

$$= \int_0^{4-2x-2y} 4[2-x] \, dz \Rightarrow \int_0^{4-2x-2y} [\boxed{8} - 4x] \, dz$$

$$= [8z]_0^{4-2x-2y} - (4xz]_0^{4-2x-2y}$$

$$= 8(4-2x-2y) - 4x[4-2x-2y]$$

$$= 32 - 16x - 16y - 16x + 8x^2 + 8xy$$

$$= 4 - 2x - 2y - 2x + x^2 + xy$$

$$= x^2 + xy - 4x - 2y + 4$$

$$= 4 //$$

Surface Integral

If $\bar{F} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ then the surface integral of $\bar{F}$ is defined as $\iint_S \bar{F}\cdot\hat{n}\,ds$ where $\hat{n}$ = outward unit normal vector to the surface S.

* In xy plane $ds = \dfrac{dx\,dy}{\hat{n}\cdot\bar{k}}$
* In yz plane $ds = \dfrac{dy\,dz}{\hat{n}\cdot\bar{i}}$
* In xz plane $ds = \dfrac{dx\cdot dz}{\hat{n}\cdot\bar{j}}$

→ Evaluate $\iint_S \bar{F}\cdot\hat{n}\,ds$ where $\bar{F} = 18z\bar{i} - 12\bar{j} + 3y\bar{k}$, $S: 2x+3y+6z=12$ in Ist octant.

Sol: $I = \iint_S \bar{F}\cdot\hat{n}\,ds$

$\hat{n} = \dfrac{\bar{n}}{|\bar{n}|}$ → outward unit normal vector

$\phi = 2x+3y+6z-12$

$$\bar{n} = \nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$= \bar{i}\frac{\partial}{\partial x}(2x+3y+6z-12) + \bar{j}\frac{\partial}{\partial y}(2x+3y+6z-12) + \bar{k}\frac{\partial}{\partial z}(2x+3y+6z-12)$$

$$= \bar{i}[2] + \bar{j}[3] + \bar{k}[6] \Rightarrow 2\bar{i}+3\bar{j}+6\bar{k}$$

$$\hat{n} = \frac{2\bar{i}+3\bar{j}+6\bar{k}}{\sqrt{4+9+36}} = \frac{2\bar{i}+3\bar{j}+6\bar{k}}{7}$$

$\bar{F} = 18z\,\bar{i} - 12\bar{j} + 3y\,\bar{k}$

$\bar{F}\cdot\hat{n} = (18z\,\bar{i} - 12\bar{j} + 3y\bar{k})\cdot\dfrac{(2\bar{i}+3\bar{j}+6\bar{k})}{7}$

$\bar{F}\cdot\hat{n} = \dfrac{1}{7}\left[36z - 36 + 18y\right]$

$ds :- \dfrac{dx\,dy}{\hat{n}\cdot\bar{k}} = \dfrac{dx\,dy}{\left(\dfrac{2\bar{i}+3\bar{j}+6\bar{k}}{7}\right)\cdot(\bar{k})} = \dfrac{7}{6}\,dx\,dy$

$I = \iint\limits_S \bar{F}\cdot\hat{n}\,ds = \iint \dfrac{1}{\cancel{7}}\left[36z - 36 + 18y\right]\cdot\dfrac{\cancel{7}}{6}\,dx\,dy$

$= \dfrac{\cancel{18}^{3}}{\cancel{6}}\iint\limits_S (2z - 2 + y)\,dx\,dy$

$= 3\iint\limits_S (y-2)\,dx\,dy$

In xy plane $z=0$. $2x+3y+6z=12$

$2x+3y=12$

$3y = 12-2x$

$y = \dfrac{12-2x}{3}$

$y:0 \Rightarrow 2x+3y=12$

$3y = 12-2x$

$y = 1-2$

$2x+3y=12$

$2x=12$

$\boxed{x=6}$

$y: 0 \to \dfrac{12-2x}{3}$

$x: 0 \to 6$

$= 3\int\limits_{x=0}^{6}\int\limits_{y=0}^{\frac{12-2x}{3}} (y-2)\,dy\,dx \Rightarrow 3\int\limits_{x=0}^{6}\left(\left[\dfrac{y^2}{2}\right]_0^{\frac{12-2x}{3}} - 2\left[y\right]_0^{\frac{12-2x}{3}}\right)dx$

$$\frac{1}{2}\int_{x=0}^{6}\cancel{\left(\frac{12-2x}{3}\right)^2}\ 144+ \qquad z=\left(\frac{12-2x-3y}{6}\right)$$

$$3\iint_S \cancel{6}\left(\frac{12-2x-\cancel{3y}}{\cancel{6}\ 3}\right)-z+y)\,dx\,dy$$

$$=3\iint_S (12-2x\cancel{-3y}-6+\cancel{3y})\,dx\,dy$$

$$=3\iint_S (6-2x)\,dx\,dy = 3\int_{x=0}^{6}\int_{y=0}^{\frac{12-2x}{3}}(6-2x)\,dy\,dx$$

$$=3\int_{x=0}^{6}\left[6y\right]_0^{\frac{12-2x}{3}}-2x\left[y\right]_0^{\frac{12-2x}{3}} \Rightarrow 3\int_{x=0}^{6}\cancel{6}^{2}\left(\frac{12-2x}{\cancel{3}}\right)-2x\left(\frac{12-2x}{3}\right)$$

$$=3\int_{x=0}^{6}\left[(24-8x)-\frac{24x+4x^2}{3}\right] \Rightarrow \cancel{3}\int_0^6 \frac{72-24x-24x+4x^2}{\cancel{3}}$$

$$\int_0^6 72-48x+4x^2 \Rightarrow \int_0^6 \left[72-44x\right]dx$$

$$\cancel{72\left[x\right]_0^6-\cancel{44}^{22}\left[\frac{x^2}{\cancel{2}}\right]_0^6 \Rightarrow 72\times 6-36\times 22}$$

$$=512-792$$

$$72[x]_0^6 - \cancel{48}^{24}\left[\frac{x^2}{\cancel{2}}\right]_0^6 + 4\left[\frac{x^3}{3}\right]_0^6$$

$$= 72\times 6 - 24\times 36 + 4\times\frac{216}{3}$$

$$= 512 - 864 + \frac{864}{3}$$

<u>Stoke's Theorem</u>

$\bar{F} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ then $\int_C \bar{F}\cdot d\bar{r} = \iint \text{curl}\,\bar{F}\cdot\hat{n}\,ds$

where S is ~~bounded~~ closed surface bounded by closed curve 'C'.

Q) $\int_C \bar{F}\cdot d\bar{r}$, where $\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$

'C' is the boundary of rectangle

$x = \pm a,\ y = 0,\ y = b$

$$\text{curl}\,\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & 0 \end{vmatrix}$$

$$= \bar{i}[-0] + \bar{j}[0] + \bar{k}[-2y-2y]$$

$$= -4y\bar{k}$$

$\boxed{\hat{n} = \bar{k}}$ ⇒ Outward unit normal vector in xy plane.

$$ds = \frac{dx\,dy}{\hat{n}\cdot\bar{k}} = \frac{dx\,dy}{\bar{k}\cdot\bar{k}} = dx\,dy$$

$$I = \int_C \bar{F} \cdot d\bar{r} = \iint -4y\bar{k} \cdot \bar{k} \Rightarrow \iint_S -4y\,dx\,dy$$

$$= -4 \iint_S y\,dx\,dy \quad x = \pm a,\ y = 0,\ y = b$$

$$= -4 \int_{x=-a}^{a} \left[\int_{y=0}^{b} y\,dy\right] dx \Rightarrow -4 \int_{x=-a}^{a} \left[\frac{y^2}{2}\right]_{y=0}^{b} dx$$

$$= -2 \int_{x=-a}^{a} (b^2 - 0)\,dx \Rightarrow -2b^2 \int_{x=-a}^{a} 1\,dx = -2b^2 [x]_{-a}^{a}$$

$$= -2b^2 [a - (-a)]$$

$$= -4ab^2$$

(1) Green's Theorem

$\int_C \bar{F} \cdot d\bar{r}$, $\bar{F} = (x^2 + y^2)\bar{i} - 2xy\bar{j}$, $y = 0, y = b$

$C \Rightarrow x = \pm a$

(2) If $\bar{F} = x^2 y^2 \bar{i} + y\bar{j}$ & the curve 'C' is $y^2 = 4x$ in xy-plane from (0,0) to (4,4)

(3) By change of order of integration

$$\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy\,dx.$$

(4) Find area in between $r = 2\cos\theta$ & $r = 4\cos$

1) A) $\bar{F} = f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$ $d\bar{r} = \bar{i}\,dx + \bar{j}\,dy + \bar{k}\,dz$

$f_1 = (x^2+y^2) \Rightarrow \frac{\partial f_1}{\partial x} = \frac{\partial}{\partial x}(x^2+y^2)$

$f_2 = -2xy$

$\int_C \bar{F}\cdot d\bar{r} = \int_C f_1\cdot dx + f_2\,dy \Rightarrow \int_C \underbrace{(x^2+y^2)}_{m}dx - \underbrace{2xy}_{n}\,dy$

$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(-2xy) = -2y \;\Big|\; \frac{\partial m}{\partial y} = \frac{\partial}{\partial y}(x^2+y^2) = 2y.$

$\iint_C [(-2y)-(2y)]\,dx\,dy$

$= \int_{x=-a}^{a}\int_{y=0}^{b} -4y\,dy\,dx \Rightarrow \int_{x=-a}^{a} -\cancel{4}^{\,2}\left[\frac{y^2}{\cancel{2}}\right]_0^b dx$

$= -2\int_{-a}^{a} b^2\,dx \Rightarrow -2\,b^2[x]_{-a}^{a}$

$-2b^2(a-(-a))$

$-2b^2(2a) = -4ab^2$ //

2) A) $\bar{F} = x^2y^2\bar{i} + y\bar{j}$

$\int_C \bar{F}\cdot d\bar{r} = \int_C \underbrace{x^2y^2}_{m}dx + \underbrace{y}_{N}dy$ $dx\,dy$

$= \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(y) = 0 \;\Big|\; \frac{\partial m}{\partial y} = \frac{\partial}{\partial y}(x^2y^2) = 2yx^2.$

$\iint_R (y - 2x^2y)\,dx\,dy.$ $x: 0 \to 4$, $y: 0 \to 4$